Spark Streaming(Exercise)

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Beginner Level

1. What is Structured Streaming, and how does it differ from traditional Spark batch processing?
2. What is a Trigger in Structured Streaming? Name the four types.
3. Explain the difference between Append, Update, and Complete output modes.
4. What is the role of checkpointing in Structured Streaming?
5. Name three common input sources supported by Spark Structured Streaming.
6. In a Spark streaming job, when does actual computation occur?

Intermediate Level

1. Explain why watermarking is required in stream-stream joins.
2. What are the limitations of stream-stream joins compared to stream-static joins?
3. How does Spark ensure exactly-once semantics in Structured Streaming?
4. Compare processing time vs event time in Structured Streaming.
5. Why should static reference data in a stream-static join fit into memory?
6. What happens if checkpointing is not configured in a streaming application?
7. Explain why Structured Streaming uses micro-batch execution.
8. Name real-world scenarios where Update mode is more suitable than Append mode.

Advanced Level

1. Explain the lifecycle of a Structured Streaming application from input source to sink.
2. Discuss the role of watermarking in preventing state explosion in stream-stream joins.
3. How does Spark Structured Streaming handle late-arriving events?
4. Why is a time-based join condition mandatory in stream-stream joins?
5. Compare stream processing with micro-batch (Spark) vs continuous processing engines (e.g., Flink).
6. What are the trade-offs between Append and Complete modes in terms of performance and correctness?
7. In Structured Streaming, how would you debug an application that processes significantly delayed events?
8. Explain the difference between checkpointing and write-ahead logs in maintaining state.